UNIT – V : INTERMEDIATE SQL

1. INDEX

An Index is a database structure that stores selected columns in an organized way so that SQL can find data faster instead of searching the entire table.

Advantages:

- Improves query/search performance
- Helps retrieve required records quickly
- Useful especially for large tables and complex SQL operations.

Disadvantages

- Indexes require extra disk space
- Frequent data insertion/updating can make index maintenance slower
- Too many indexes may reduce writing performance.

2. INDEX SYNTAX

An index can be added to a particular column and removed using its index name.

```
CREATE INDEX index_name
ON table_name(column_name);
```

Remove:

```
DROP INDEX index_name;
```

The PDF emphasizes that the index name identifies the index, while the column is the data being indexed.

3. TYPES OF INDEX

Type	Important Point
INDEX	Default/general index category in MySQL
PRIMARY KEY	Unique key; an index is automatically created
UNIQUE INDEX	Prevents duplicate values
FULL TEXT	Used mainly for keyword/text searching

Primary-key indexes can also improve the performance of **JOIN** operations, especially when tables are connected using IDs.

4. INDEX BEST PRACTICES

Remember these 6 principles:

1. Prefer UNIQUE INDEX, otherwise use KEY.
2. Index columns frequently used in operations such as GROUP BY.
3. Index frequently searched WHERE columns with sufficiently different values.
4. Remember that indexes consume disk space.
5. Prefer indexing columns with relatively small values.
6. Avoid indexing columns containing NULL where possible because it may affect efficiency.

5. DATABASE USER MANAGEMENT

SQL also provides features to create and manage database users and control their permissions, which is important for database security.

CREATE USER

Used to create a new database user.

```
CREATE USER 'john'@'localhost'
IDENTIFIED BY 'password';
```

Three important parts:

- `CREATE USER` → creates the account
- `'john'@'localhost'` → username + connection location
- `IDENTIFIED BY` → sets the password

6. NETWORK CONTEXT

The network context specifies where a user can connect from.

- localhost → only from the same machine
- Specific IP → only from that particular IP
- % → connection allowed from anywhere; should be used carefully because it can increase security risk.

7. SHOW USER

Used to view database users and their privilege information.

```
SELECT User, Host
FROM mysql.user;
```

The PDF shows user information such as Host, User and privileges.

8. GRANT PRIVILEGES

After creating a user, permissions can be assigned using GRANT.

```
GRANT SELECT
ON database_name.*
TO 'john'@'localhost';
```